

ASSIGNMENT 1
DIGITAL FUNDAMENTALS (BE03000071)
DEPARTMENT OF COMPUTER ENGINEERING
DIVISIONS E, F

- 1) Perform the following conversions.
 - a. $(11.10)_{10}$ in to Hexadecimal
 - b. $(ABD.AB)_{16}$ into Binary
 - c. $(625.43)_8$ into Decimal number
- 2) (a) Perform Binary subtraction using 1's complement method.
 $(10110111)_2 - (11001110)_2$
(b) Perform Binary subtraction using 2's complement method
 $(1110\ 1110)_2 - (0101\ 0101)_2$
- 3) Explain the characteristics of Digital ICs.
- 4) Explain how Hamming code is useful for correcting and detecting the errors.
- 5) Give the classification of logic families. Compare CMOS and TTL logic families.
- 6) Minimize the following Boolean function using K-map:
 $F(A,B,C,D) = \Sigma m(1, 5, 6, 12, 13, 14) + d(2, 4)$
- 7) Implement the following Boolean Function using 8:1 multiplexer.
 $F(A,B,C,D) = \Sigma m(2,3,5,7,8,9,12,13,14,15)$
- 8) Explain Quine McCluskey method of simplification of the Boolean function.
- 9) Design a Binary to Gray Code converter.
- 10) Design a two bit comparator using logic gates.

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General Instructions:

- 1) The assignment should be submitted on or before 21.09.26.
- 2) It is to be submitted in hard copy.
- 3) This assignment should not be uploaded to the PBL Task Management System. Only PBL Assignments are to be uploaded to the PBL portal.